

MERCURY

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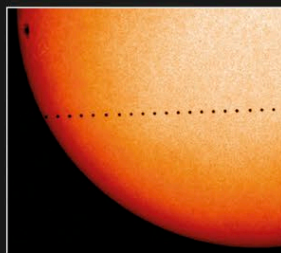
MERCURY IS THE SMALLEST planet in the solar system, the closest planet to the Sun, and the richest in iron. The surface environment is extremely harsh. There is hardly any shielding atmosphere, and the temperature rises to a blistering 800°F (430°C) during the day, then plummets

to an air-freezing –290°F (–180°C) at night. No other planet experiences such a wide range of temperatures. Its surface has been churned up by meteoritic bombardment and is dark and dusty.

ORBIT

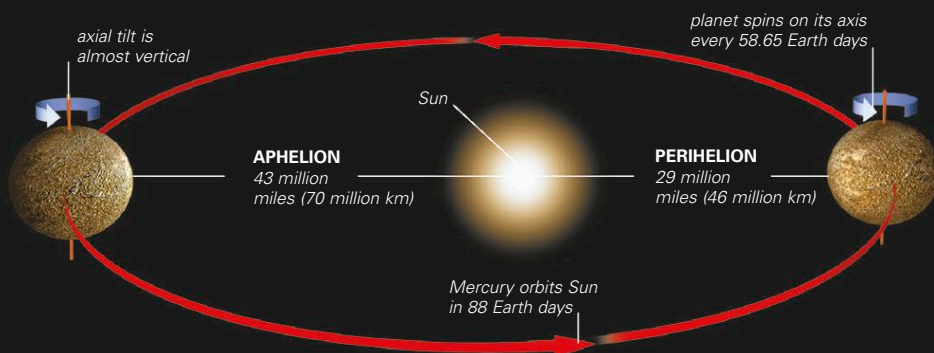
With the exception of Pluto, Mercury has the most eccentric of all the planetary orbits. At perihelion, it is only 28.6 million miles (46 million km) from the Sun, but at aphelion, it is 43.3 million miles (69.8 million km) away. The plane of Mercury's equator coincides with the plane of its orbit. (In other words, its axis of rotation is almost vertical.) This means that the planet has no seasons and that some craters close to the poles never receive any sunlight and are permanently cold. The orbit is inclined at 7° to the plane of the Earth's orbit. Because Mercury orbits inside the Earth's orbit, it displays phases, just like the Moon (see p.62).

TRANSIT OF MERCURY
Mercury passes directly between Earth and the Sun about 13 times a century. This row of dots is a multiple exposure of Mercury's transit across the Sun in 2006.



SPIN AND ORBIT

Mercury rotates three times in two orbits. (In other words, there are 3 Mercurian "days" in 2 Mercurian "years.") This unusual spin-orbit coupling means that for an observer standing on Mercury, there would be an interval of 176 Earth days between one sunrise and the next.



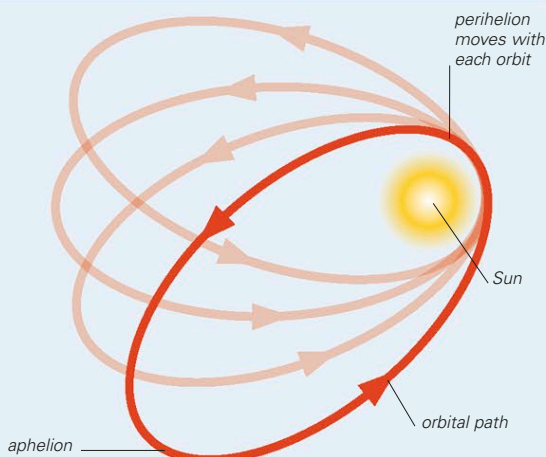
EXPLORING SPACE

EINSTEIN AND MERCURY

Mercury's perihelion position moves slightly more than Isaac Newton's theories of motion predict. In the 19th century, it was proposed that a planet (called Vulcan) inside Mercury's orbit produced this effect. In his general theory of relativity of 1915, the German physicist Albert Einstein suggested that space near the Sun was curved and correctly predicted the exact amount by which the perihelion would move.

MERCURY'S WOBBLY ORBIT

Mercury's perihelion advances by about 1.55° every century, which is 0.012° more than is expected given the gravitational influence of nearby planets.



POCKMARKED PLANET

Mercury's heavily cratered surface, seen here from the MESSENGER probe in 2009, resembles the highland areas of the Moon. The planet also has large expanses of younger, smooth, lightly cratered plains, similar to the lunar maria.